

Machine-dependent nonlinear finite element responses in computational shell buckling: a crowdsourced Abaqus benchmark

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Abstract

This short communication reports on a crowdsourced benchmark performed in 2026 that demonstrates macroscopically different non-linear equilibrium path families arising from serial execution of identical Abaqus input files on different machines and installations. The benchmark is a cylindrical shell under uniform meridional compression, with a control case (plastic collapse) that was expected to be insensitive to the execution platform and an interesting case (elastic buckling) which was expected to be highly platform sensitive. Three dominant response families were found in the dataset, aligned with CPU vendor and Abaqus release, of which only one follows the expected fundamental path for the highlighted nonlinear elastic buckling case over the computed range. The variability when quantified using pairwise Fréchet distances appears genuine, but the crowdsourced dataset is too small and self-selected to support a causal attribution. The findings raise questions about the reproducibility of nonlinear structural predictions near unstable equilibrium paths. We release the benchmark and anonymised data in the hope of starting a discussion and facilitating a larger controlled cross-platform and cross-solver experiment covering additional benchmarks at a future time.

Keywords: computational reproducibility, nonlinear finite element analysis, shell buckling, crowdsourced benchmark, Fréchet distance, cluster analysis, Abaqus

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1. The benchmark

It all started with a simple task, to execute an Abaqus input file unchanged on a different machine. The task was part of an undergraduate assignment in computational shell buckling at Imperial College London, where students were modelling a cylindrical shell under uniform meridional compression using increasingly sophisticated finite element analyses. An early task was a materially nonlinear but geometrically linear analysis (MNA) under an elastic-perfectly plastic material model where a distributed edge load per unit circumference of magnitude equal to the plastic collapse load $N_y = \sigma_y t$ was applied (σ_y is a yield stress, t is a unit thickness). This was to give an initial linear path followed by an indefinite plateau at a normalised load proportionality factor of unity. A collapse mechanism should form first through plastic hinging in the boundary layers, followed by the spread of membrane yielding through the rest of the shell (Fig. 1, left). A later task was an unperturbed geometrically nonlinear but elastic analysis (GNA) under an edge load of magnitude equal to the classical elastic critical buckling load

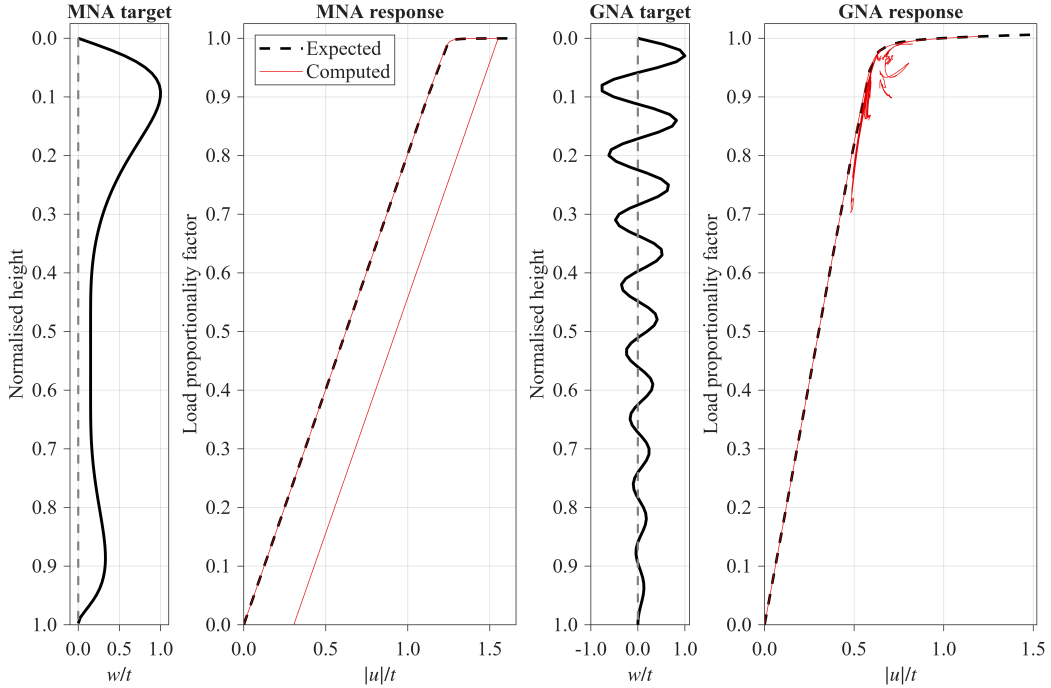


Figure 1: Target deformation profiles and corresponding response paths for the materially nonlinear analysis (MNA) and geometrically nonlinear analysis (GNA) benchmark cases. The ‘target’ panels show the normalised radial deformation w/t over the normalised height, ‘response’ panels show the load proportionality factor against the normalised loaded-edge meridional shortening $|u|/t$. Here t is a unit thickness.

$N_{cl} = \sigma_{cl}t \approx 0.605Et^2/r$ (assuming a Poisson ratio of 0.3, E is a Young's modulus and r is a midsurface radius), intended to trace the fundamental equilibrium path of this system. Less obviously and much less universally appreciated (Section 9.5a in [2]), this response should likewise form a plateau at a load proportionality factor of unity as the bending boundary layers grow without bound and fully elastic sinusoidal deformations penetrate into the shell (Fig. 1, right). No formal geometric imperfections of any kind were introduced. Things got interesting when students obtained neither expected response consistently, but instead obtained mostly non-physical results that varied between machines.

The idea of a systematic reproducibility benchmark crystallised from this experience, and a public call to crowdsource a larger dataset was made. The benchmark consisted of four small, near-identical Abaqus input files. One was an MNA 'control' which was not intended to be interesting (i.e. it was expected to give the same macroscopic response across all execution platforms), and three GNA files differing only by the initial and maximum arc-length increment sizes. Participants were asked to run all files to completion, serially and without modification, then execute a Python script to extract loading histories, and document the hardware (CPU, memory) and software (Abaqus build, OS) environments in a metadata questionnaire.

2. Results

A total of 33 submissions were received whose distinct load histories are illustrated in Fig. 1. All MNAs exhibit the correct initial linear response, briefly follow a plateau approaching the expected load level but then erroneously reverse the direction of loading. This reversal is a manifestation of the 'doubling back' pathology [3] known to occur in arc-length nonlinear solvers, but at least it was obtained consistently by all submissions. The GNAs, however, despite following the correct initial near-linear response eventually diverged into multiple macroscopically different late-stage paths as highlighted in Fig. 2, *most* of them incorrectly departing from the expected plateauing fundamental path. The fundamental GNA equilibrium becomes energetically unstable after the first bifurcation point, which for the benchmark's boundary conditions should occur at a normalised load level of approximately 0.9 (Figure 2.44 in [5] for $Z \approx 10^3$, C1-S3). Absent an explicit perturbation of the initial geometry (the standard technique for triggering bifurcations by transforming them into very sharp limit points), or some other source of perturbation, the solver should be expected to be able to trace this unstable but smooth fundamental path correctly.

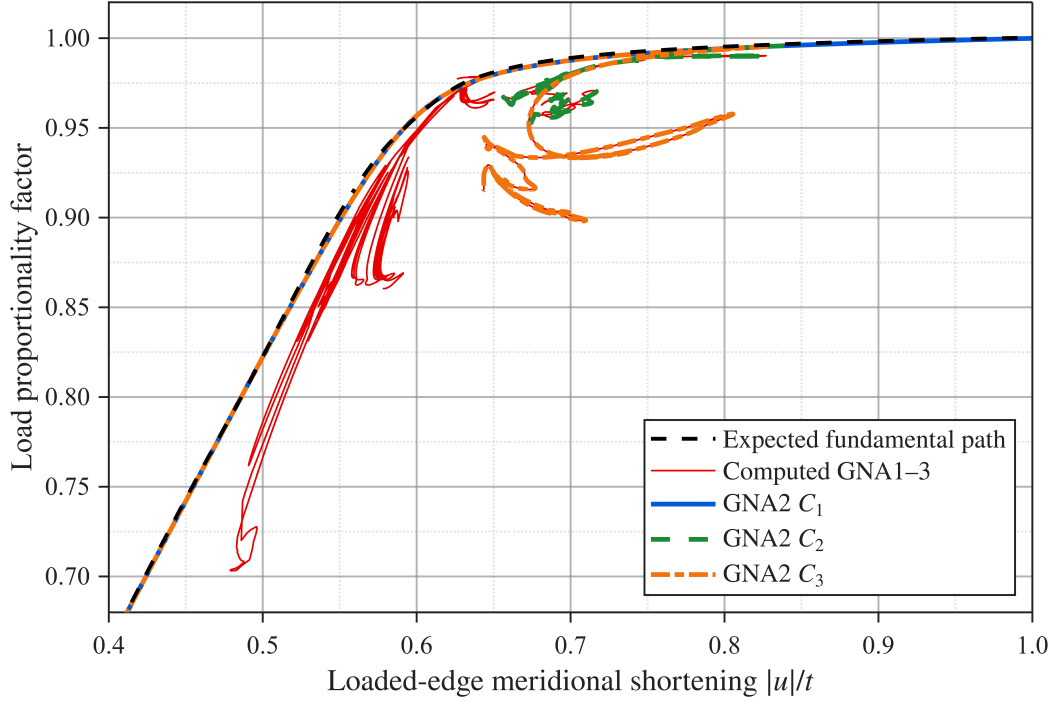


Figure 2: Enlarged view of the GNA response paths near the expected plateau. Three dominant GNA2 response families C_1 – C_3 are highlighted with colours matching the dendrogram in Fig. 4. The dashed black curve shows the expected fundamental equilibrium path; only C_1 follows it over the computed range. Loaded-edge meridional shortening is normalised by the shell thickness t .

55 Deterministic SHA256 fingerprints identified 12, 11, 14 and 11 unique
 56 histories for MNA, GNA1, GNA2 and GNA3, respectively. Pairwise mul-
 57 tivariate Fréchet distances [4] (which preserve the ordering of points along
 58 each history) were computed between these unique histories and are shown as
 59 colour-coded matrices in Fig. 3 for the MNA and GNA2 cases only. MNA his-
 60 tories, intended to be an exactly reproducible control, unexpectedly contained
 61 microscopic differences. That these still produce the same (albeit erroneous)
 62 macroscopic plastic collapse response is consistent with MNAs being pertur-
 63 bation insensitive and thus appropriate for use as a macroscopic ‘control’, for
 64 this system at least. By contrast, the GNA responses exhibit severe variabil-
 65 ity, with microscopic differences cascading to give macroscopically radically
 66 different histories, apparently from platform-specific numerical causes. This
 67 is consistent with the known sensitivity of thin cylinders under meridional
 68 compression to imperfections (Sections 4.6 & 4.7 in [5]), and generally with
 69 the sensitivity of unstable nonlinear systems to perturbation.

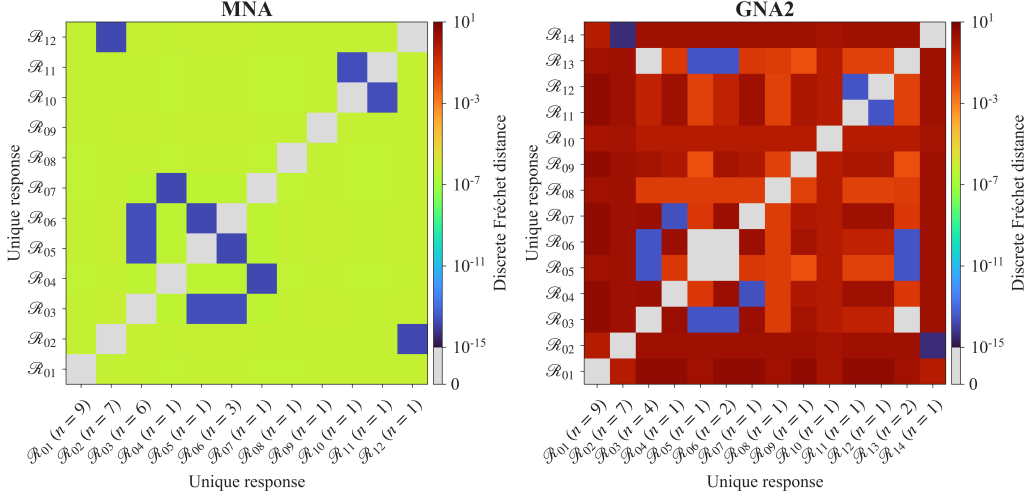


Figure 3: Pairwise discrete Fréchet distances between the unique MNA and GNA2 response histories represented by the standardised load proportionality factor, loaded-edge meridional shortening and maximum force residual, with path ordering preserved. Each coordinate is scaled by its standard deviation across the concatenated unique histories for that analysis; missing residuals are omitted from pointwise comparisons. Colours show distance on a common logarithmic scale, grey denoting zero. n is the number of submissions for each unique response $\mathcal{R}_i$. At least part of the microscopic variability may still be attributable to export-formatting effects.

70 Recurring GNA paths indicate structured, partly reproducible variability.
71 Complete-linkage clustering, which favours internally compact clusters [1],
72 performed on the basis of the pairwise Fréchet distances between response
73 histories (i.e. *not* on the basis of additional questionnaire metadata) suggests
74 three dominant families of submissions in the dataset. These are Intel CPUs
75 with older Abaqus versions 2020-23 (C_1 , nine submissions), AMD CPUs with
76 newer Abaqus versions 2024-25 (C_2 , nine submissions) and Intel CPUs with
77 newer Abaqus versions 2024-25 (C_3 , eight submissions). The remaining seven
78 submissions are, in this dataset, distributed across three singleton and two
79 doubleton clusters. These GNA2 clusters are illustrated in Fig. 4 and it
80 may be noted that they remain unchanged across several order of magnitude
81 of Fréchet distance threshold, indicating a persistent separation between
82 response families. Of these, only C_1 for GNA2 actually follows the expected
83 response (Fig. 2). The memberships of the AMD and newer-Intel dominant
84 clusters remain unchanged across MNA, GNA1 and GNA3, while the older-
85 Intel dominant cluster varies only slightly. This recurrence of clusters across
86 the four analyses is consistent with a platform-dependent hypothesis, but the
87 small, self-selected dataset does not separate the effects of hardware, software
88 release and other execution-environment differences sufficiently to support a
89 causal attribution.

105 • The system under investigation, a thin cylindrical shell under uniform
 106 meridional compression, is very well understood and both MNA and
 107 GNA benchmarks have known analytical solutions. This made identifying
 108 incorrect responses straightforward. For less well-understood
 109 nonlinear structural systems, such solver- or platform-dependent pathologies
 110 may be much harder to identify, let alone resolve, and the present
 111 findings raise questions about just how reproducible nonlinear structural
 112 predictions are in general.

113 The authors call for a larger controlled cross-platform and cross-solver re-
 114 producibility study on more safety-critical structural systems of industrial
 115 significance to raise awareness and accurately identify correlations, causal
 116 mechanisms and countermeasures. A database of ‘sensitive’ benchmarks such
 117 as this one would be a particularly valuable resource for the community.

118 Declarations

119 **CRedit authorship statement.** Yuhan Wei: Formal analysis, Writing - original
 120 draft. Molly Balassa: Conceptualization, Formal analysis, Writing - original draft.
 121 Adam J. Sadowski: Conceptualization, Formal analysis, Writing - original draft.

122 **Data and code availability.** Anonymised data, benchmark materials and pro-
 123 cessing code are available at https://github.com/SadowskiAJ/RRX_public.git.
 124 There is significantly more in the data than has been reported on here.

125 **Declaration of generative AI use.** OpenAI Codex was used to assist with
 126 development of the analysis code. All resulting code was reviewed by the authors.

127 **Competing interests.** The authors declare no competing interests.

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